

Catalan numbers, shuffle theorems, and Macdonald polynomials

George H. Seelinger

joint work with J. Blasiak, M. Haiman, J. Morse, and A. Pun

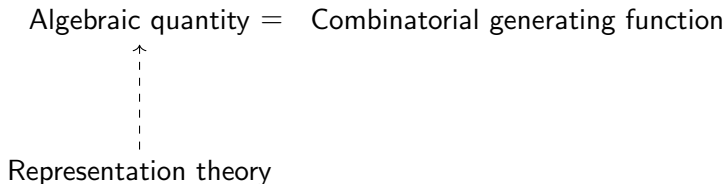
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SMRI: Modern Perspectives in Representation Theory

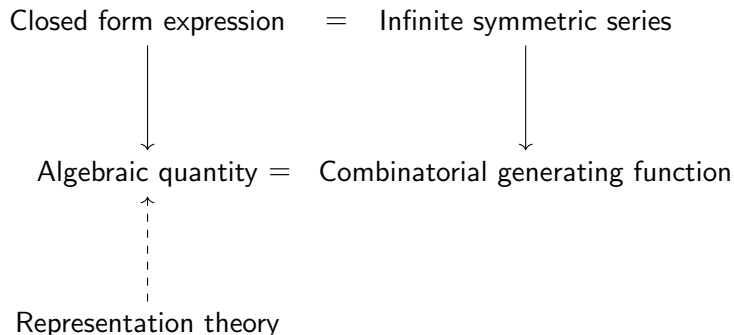
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Algebraic quantity = Combinatorial generating function

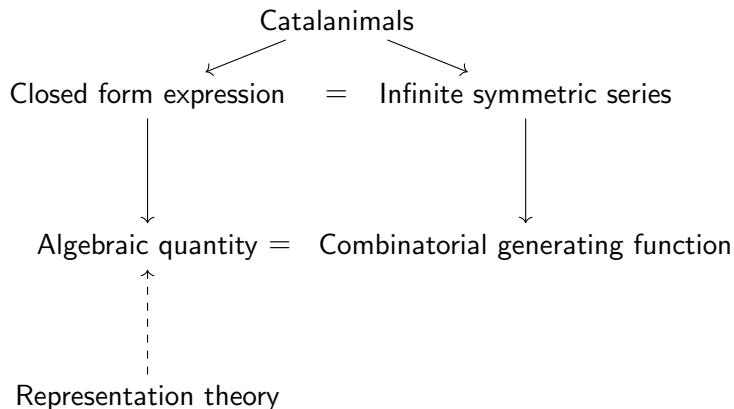
Overview



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- ① **Background on symmetric functions and Macdonald polynomials**
- ② Shuffle theorems, combinatorics, and LLT polynomials
- ③ A new formula for Macdonald polynomials

Symmetric polynomials and functions

- Symmetric polynomials $\mathbb{Q}[z_1, \dots, z_n]^{S_n}$
- Generators

$$e_r(z_1, \dots, z_n) = \sum_{1 \leq i_1 < i_2 < \dots < i_r \leq n} z_{i_1} \cdots z_{i_r}$$

$$e_1(z_1, z_2, z_3) = z_1 + z_2 + z_3$$

$$e_2(z_1, z_2, z_3) = z_1 z_2 + z_1 z_3 + z_2 z_3$$

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- Degree d elements of $\Lambda = \mathbb{Q}$ -vector space of dimension. . . ?

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- Integer partitions of d .

Partitions

Definition

$n \in \mathbb{Z}_{>0}$, a *partition of n* is $\lambda = (\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_\ell > 0)$ such that $\lambda_1 + \lambda_2 + \cdots + \lambda_\ell = n$.

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$$5 \rightarrow \begin{array}{|c|c|c|c|c|}\hline\hline\hline\hline\hline\hline\end{array}$$

$$4 + 1 \rightarrow \begin{array}{|c|c|c|c|c|}\hline\hline\hline\hline\hline\end{array}$$

$$3 + 2 \rightarrow \begin{array}{|c|c|c|c|}\hline\hline\hline\hline\hline\end{array}$$

$$3 + 1 + 1 \rightarrow \begin{array}{|c|c|c|c|}\hline\hline\hline\hline\hline\end{array}$$

$$2 + 2 + 1 \rightarrow \begin{array}{|c|c|c|}\hline\hline\hline\hline\hline\end{array}$$

$$2 + 1 + 1 + 1 \rightarrow \begin{array}{|c|c|c|c|}\hline\hline\hline\hline\hline\end{array}$$

$$1 + 1 + 1 + 1 + 1 \rightarrow \begin{array}{|c|c|c|c|c|}\hline\hline\hline\hline\hline\hline\end{array}$$

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Filling of partition diagram of λ with numbers such that

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Collection is called $\text{SSYT}(\lambda)$.

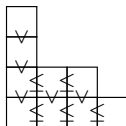
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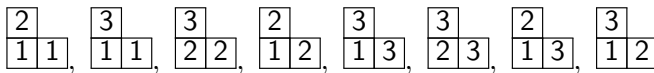
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For $\lambda = (2, 1)$,



Polynomials from tableaux

Associate a polynomial to $\text{SSYT}(\lambda)$.

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2		3		3		2		3		3		2		3	
1	1	1	1	2	2	1	2	1	3	2	3	1	3	1	2

Polynomials from tableaux

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$$s_{(2,1)}(z_1, z_2, z_3) = z_1^2 z_2 + z_1^2 z_3 + z_2^2 z_3 + z_1 z_2^2 + z_1 z_3^2 + z_2 z_3^2 + 2z_1 z_2 z_3$$

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$$s_\lambda = \sum_{T \in \text{SSYT}(\lambda)} \mathbf{z}^T \text{ for } \mathbf{z}^T = \prod_{i \in T} z_i$$

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- s_λ is a symmetric function.
- $\{s_\lambda\}_\lambda$ forms a basis for Λ .

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Hidden Guide: Schur Positivity

“Naturally occurring” symmetric functions which are non-negative (coefficients in $\mathbb{N}$) linear combinations in Schur polynomial basis are interesting since they could have representation-theoretic models.

A Graded Example

$$\Delta = \det \begin{vmatrix} x_1^2 & x_1 & 1 \\ x_2^2 & x_2 & 1 \\ x_3^2 & x_3 & 1 \end{vmatrix} = x_1^2(x_2 - x_3) - x_2^2(x_1 - x_3) + x_3^2(x_1 - x_2)$$

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$S_3 \curvearrowright \text{span}\{\Delta\}$ via $\sigma.\Delta = \text{sgn}(\sigma)\Delta$.

Consider $S_3 \curvearrowright M = \text{all partial derivatives of } \Delta$.

$$\begin{array}{c} \text{span}\{\Delta\} \\ \downarrow \partial_{x_i} \\ \text{span}\{2x_1(x_2 - x_3) - x_2^2 + x_3^2, 2x_2(x_3 - x_1) - x_3^2 + x_1^2\} \\ \downarrow \partial_{x_i} \\ \text{span}\{x_3 - x_1, x_2 - x_3\} \\ \downarrow \partial_{x_i} \\ \text{span}\{1\} \end{array}$$

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 \text{span}\{\Delta\} & & \cong V_{\begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \square \\ \hline \end{array}} \\
 \downarrow \partial_{x_i} & & \\
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 \downarrow \partial_{x_i} & & \\
 \text{span}\{x_3 - x_1, x_2 - x_3\} & & \cong V_{\begin{array}{|c|} \hline \square \square \\ \hline \square \square \\ \hline \end{array}} \\
 \downarrow \partial_{x_i} & & \\
 \text{span}\{1\} & & \cong V_{\begin{array}{|c|} \hline \square \square \square \\ \hline \end{array}}
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 \text{span}\{1\} & & \cong V_{\begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \square \\ \hline \end{array}}
 \end{array}$$

- $\text{Frob}(M) = s_{\begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \square \\ \hline \end{array}} + 2s_{\begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array}} + s_{\begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \square \\ \hline \end{array}}$

A Graded Example

$$\begin{array}{ccc}
 \text{span}\{\Delta\} & & \cong V_{\begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \square \\ \hline \end{array}} \\
 \downarrow \partial_{x_i} & & \\
 \text{span}\{2x_1(x_2 - x_3) - x_2^2 + x_3^2, 2x_2(x_3 - x_1) - x_3^2 + x_1^2\} & & \cong V_{\begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array}} \\
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- Problem: what about grading?

Getting More Information

$$\begin{array}{llll} \text{span}\{\Delta\} & & \cong V_{\begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \square \\ \hline \end{array}} & \text{deg} = 3 \\ \downarrow \partial_{x_i} & & & \\ \text{span}\{2x_1(x_2 - x_3) - x_2^2 + x_3^2, 2x_2(x_3 - x_1) - x_3^2 + x_1^2\} & & \cong V_{\begin{array}{|c|} \hline \square \\ \hline \square \square \\ \hline \end{array}} & \text{deg} = 2 \\ \downarrow \partial_{x_i} & & & \\ \text{span}\{x_3 - x_1, x_2 - x_3\} & & \cong V_{\begin{array}{|c|} \hline \square \\ \hline \square \square \\ \hline \end{array}} & \text{deg} = 1 \\ \downarrow \partial_{x_i} & & & \\ \text{span}\{1\} & & \cong V_{\begin{array}{|c|} \hline \square \square \square \\ \hline \end{array}} & \text{deg} = 0 \end{array}$$

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 \downarrow \partial_{x_i} & & & \\
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 \end{array}$$

- $(V_\lambda \text{ in degree } d) \mapsto q^d s_\lambda$

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Getting More Information

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- Hall-Littlewood polynomial $H_{\begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \square \\ \hline \end{array}}(X; q).$
- Remark: $M \cong \mathbb{Z}[x_1, x_2, x_3]/\text{Sym}^+ \cong H^*(Fl_3)$ as **graded** S_3 -representations.

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$$\tilde{H}_\lambda(X; q, t) = \sum_{\mu} \tilde{K}(q, t) s_{\mu} \text{ conjecturally satisfies } \tilde{K}(q, t) \in \mathbb{N}[q, t]$$

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- $\tilde{H}_\lambda(X; 1, 1) = s_1^{|\lambda|}$.
- Does there exist a family of S_n -regular representations whose bigraded Frobenius characteristics equal $\tilde{H}_\lambda(X; q, t)$?

Garsia-Haiman modules

- $S_n \curvearrowright \mathbb{Q}[x_1, \dots, x_n, y_1, \dots, y_n]$ with $\sigma(x_i) = x_{\sigma(i)}$, $\sigma(y_j) = y_{\sigma(j)}$.

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$\tilde{H}_\lambda(X; q, t) = \sum_\mu \tilde{K}_{\lambda\mu}(q, t) s_\mu$ satisfies $\tilde{K}_{\lambda\mu}(q, t) \in \mathbb{N}[q, t]$.

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- No combinatorial description of $\tilde{K}_{\lambda\mu}(q, t)$.

Symmetric functions, representation theory, and combinatorics

Symmetric function	Representation theory	Combinatorics
s_λ	Irreducible V_λ	SSYT(λ)
$\tilde{H}_\lambda(X; q, t)$	Garsia-Haiman M_λ	Later

Observation

All of these Garsia-Haiman modules are contained in the module of diagonal harmonics:

$$DH_n = \text{span}\{f \in \mathbb{C}[x_1, \dots, x_n, y_1, \dots, y_n] \mid \left(\sum_{j=1}^n \partial_{x_j}^r \partial_{y_j}^s \right) f = 0, \forall r+s > 0\}$$

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Question

What symmetric function is the bigraded Frobenius characteristic of DH_n ?

Frobenius characteristic of DH_3

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$$= \frac{t^3 \tilde{H}_{1,1,1}}{-qt^2 + t^3 + q^2 - qt} - \frac{(q^2 t + qt^2 + qt) \tilde{H}_{2,1}}{-q^2 t^2 + q^3 + t^3 - qt} - \frac{q^3 \tilde{H}_3}{-q^3 + q^2 t + qt - t^2}$$

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Compare to

$$e_3 = \frac{\tilde{H}_{1,1,1}}{-qt^2 + t^3 + q^2 - qt} - \frac{(q+t+1) \tilde{H}_{2,1}}{-q^2 t^2 + q^3 + t^3 - qt} - \frac{\tilde{H}_3}{-q^3 + q^2 t + qt - t^2}$$

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Operator ∇

$$\nabla \tilde{H}_\lambda(X; q, t) = q^{n(\lambda)} t^{n(\lambda^*)} \tilde{H}_\lambda(X; q, t),$$

where $n(\lambda) = \sum_i (i-1)\lambda_i$ and λ^* is the transpose partition to λ .

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Theorem (Haiman, 2002)

The bigraded Frobenius characteristic of DH_n is given by ∇e_n .

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s_λ	Irreducible V_λ	SSYT(λ)
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∇e_n	DH_n	Now: Shuffle theorem

Outline

- ① Background on symmetric functions and Macdonald polynomials
- ② **Shuffle theorems, combinatorics, and LLT polynomials**
- ③ A new formula for Macdonald polynomials

A Combinatorial Connection: Shuffle Theorem

Theorem (Carlsson-Mellit, 2018)

$$\nabla e_k = \sum_{\lambda} (q, t \text{ monomial})(LLT \text{ polynomial})$$

- Summation over all k -by- k Dyck paths.

A Combinatorial Connection: Shuffle Theorem

Theorem (Carlsson-Mellit, 2018)

$$\nabla e_k = \sum_{\lambda} t^{\text{area}(\lambda)} q^{\text{dinv}(\lambda)} (LLT \text{ polynomial})$$

- Summation over all k -by- k Dyck paths.
- $\text{area}(\lambda)$ and $\text{dinv}(\lambda)$ statistics of Dyck paths.

A Combinatorial Connection: Shuffle Theorem

Theorem (Carlsson-Mellit, 2018)

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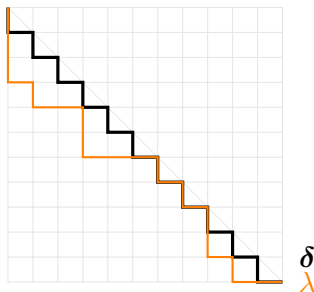
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- ω a standard involution of symmetric polynomials.
- Conjectured by (Haiman-Haglund-Loehr-Remmel-Ulyanov, 2002).

Dyck paths

Dyck paths

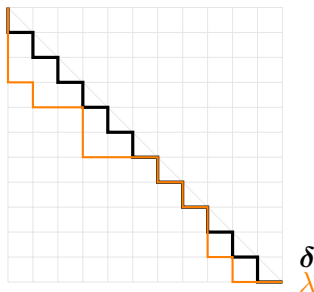
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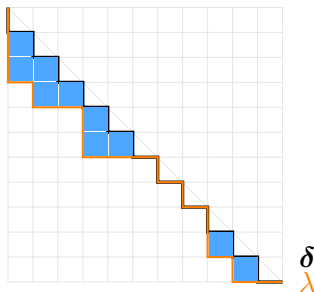


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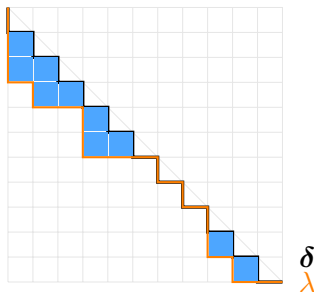


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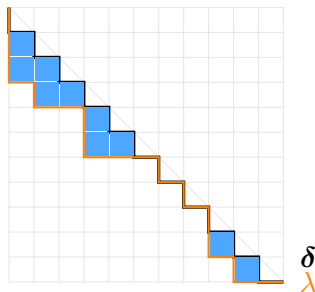


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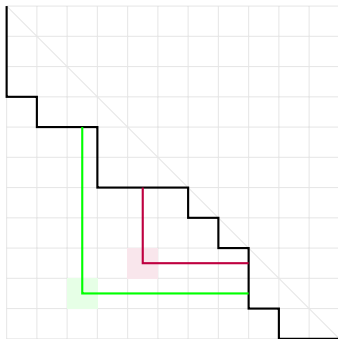
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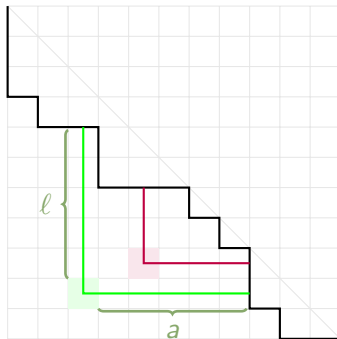
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dinv

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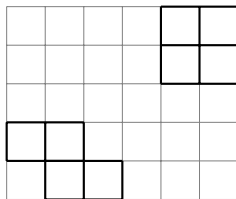
Balanced hook is given by a cell below λ satisfying

$$\frac{l}{a+1} < 1 - \epsilon < \frac{l+1}{a}, \quad \epsilon \text{ small.}$$

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-4	-3	-2	-1	0	1
-3	-2	-1	0	1	2
-2	-1	0	1	2	3
-1	0	1	2	3	4
0	1	2	3	4	5

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$T =$

				5	6
				1	1
2	4				
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- $\mathcal{G}_\nu$ is Schur-positive for any tuple of skew shapes ν [Grojnowski-Haiman, 2007].

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$$\lambda \rightarrow q^{\text{dinv}(\lambda)} t^{\text{area}(\lambda)} \rightarrow q^{\text{dinv}(\lambda)} t^{\text{area}(\lambda)} \omega \mathcal{G}_{\nu(\lambda)}(X; q^{-1})$$

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$$q^3$$



$$q^2 t$$



$$qt$$



$$qt^2$$



$$t^3$$

Example ∇e_3

λ	$q^{\text{dinv}(\lambda)} t^{\text{area}(\lambda)}$	$q^{\text{dinv}(\lambda)} t^{\text{area}(\lambda)} \omega \mathcal{G}_{\nu(\lambda)}(X; q^{-1})$
	q^3	$s_3 + qs_{2,1} + q^2 s_{2,1} + q^3 s_{1,1,1}$
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- Entire quantity is q, t -symmetric
- Coefficient of $s_{1,1,1}$ in sum is a “ (q, t) -Catalan number” $(q^3 + q^2 t + qt + qt^2 + t^3)$.

Generalizing Shuffle Theorem

When a problem is too difficult, try generalizing!

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What generalizes ∇e_k ?

Elliptic Hall Algebra

Algebra $\mathcal{E} \curvearrowright \Lambda =$ symmetric polynomials

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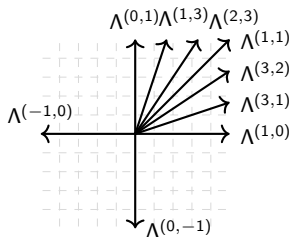
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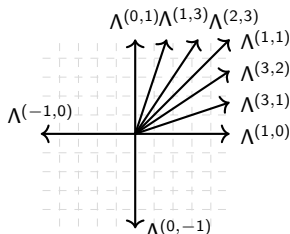


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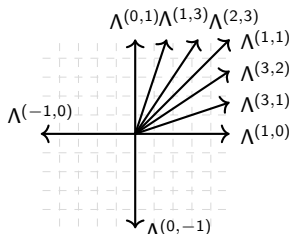
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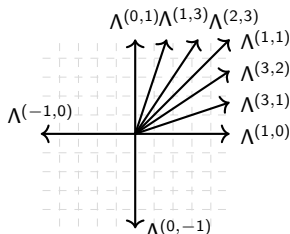
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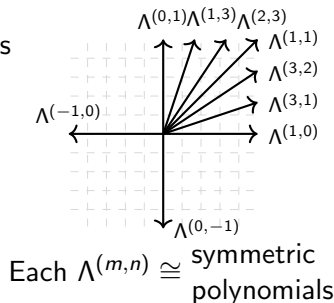


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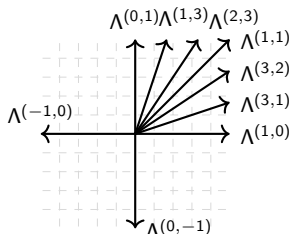
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- $e_k^{(1,1)} \cdot 1 = \nabla e_k^{(0,1)} \nabla^{-1} \cdot 1 = \nabla e_k$.
- Can be difficult to work with in general. Can we make it more explicit?

Root ideals

$R_+ = \{\alpha_{ij} \mid 1 \leq i < j \leq n\}$ denotes the set of positive roots for GL_n , where $\alpha_{ij} = \epsilon_i - \epsilon_j$.

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A root ideal $\Psi \subseteq R_+$ is an upper order ideal of positive roots.

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$\Psi = \text{Roots above Dyck path}$

Weyl symmetrization

Define the *Weyl symmetrization operator*

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Catalanimals

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$$\begin{aligned} H(R_+, R_+, \{\alpha_{13}\}, (111)) &= \text{pol}_X \sigma \left(\frac{z^{111} (1 - qt z_1/z_3)}{\prod_{1 \leq i < j \leq 3} (1 - q z_i/z_j) (1 - t z_i/z_j)} \right) \\ &= s_{111} + (q + t + q^2 + qt + t^2) s_{21} + (qt + q^3 + q^2 t + qt^2 + t^3) s_3 \\ &= \omega \nabla e_3. \end{aligned}$$

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Let $R_+ = \{\alpha_{ij} \mid 1 \leq i < j \leq l\}$ and $R_+^0 = \{\alpha_{ij} \in R_+ \mid i + 1 < j\}$.

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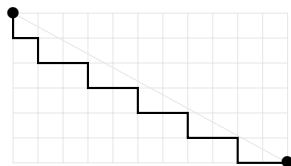
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for $\mathbf{b} = (b_0, \dots, b_{km-1})$ satisfying $b_i =$ the number of south steps on vertical line $x = i$ of highest lattice path under line $y + \frac{n}{m}x = n$.

δ = highest Dyck path.



δ

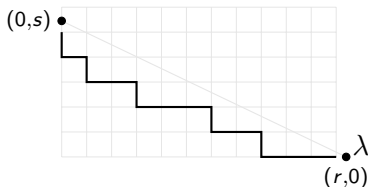
$$\mathbf{b} = (1, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0)$$

Results

Manipulating Catalanimal $\implies$ a proof of the Rational Shuffle Theorem + a generalization.

Theorem (Blasiak-Haiman-Morse-Pun-S., 2023)

Given $r, s \in \mathbb{R}_{>0}$ such that $p = s/r$ irrational, take $\mathbf{b} = (b_1, \dots, b_l) \in \mathbb{Z}^l$ to be the south step sequence of highest path δ under the line $y + px = s$.



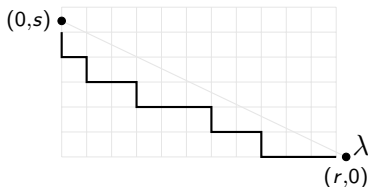
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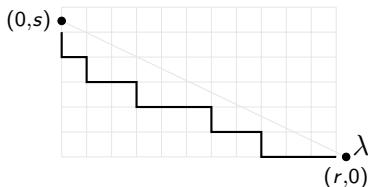
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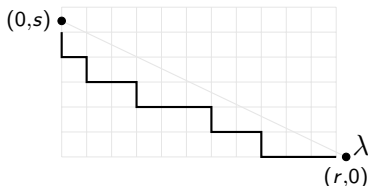
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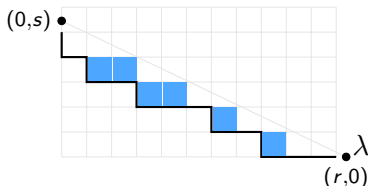
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$\text{area}(\lambda)$ as before

$\text{div}_p(\lambda) = \#p\text{-balanced hooks}$

$$\frac{\ell}{a+1} < p < \frac{\ell+1}{a}$$

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Special case: $\mathcal{G}_\nu^{(1,1)} \cdot 1 = \nabla \mathcal{G}_\nu(X; q)$.

LLT Catalanimals

For a tuple of skew shapes ν , the *LLT Catalanimal* $H_\nu = H(R_q, R_t, R_{qt}, \lambda)$ is determined by

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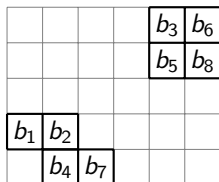
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Listing this filling in reading order gives λ .

LLT Catalanimals

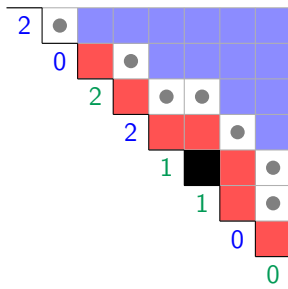
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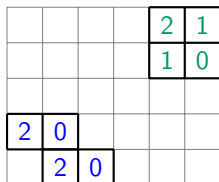


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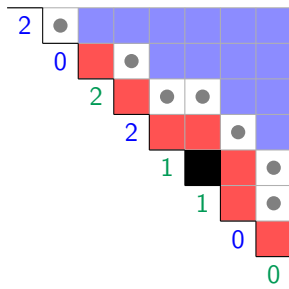
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λ , as a filling of ν



Theorem (Blasiak-Haiman-Morse-Pun-S., 2024)

Let ν be a tuple of skew shapes and let $H_\nu = H(R_q, R_t, R_{qt}, \lambda)$ be the associated LLT Catalanimal. Then

$$\begin{aligned}\nabla \mathcal{G}_\nu(X; q) &= c_\nu \omega H_\nu \\ &= c_\nu \omega \sigma \left(\frac{z^\lambda \prod_{\alpha \in R_{qt}} (1 - qt z^\alpha)}{\prod_{\alpha \in R_q} (1 - q z^\alpha) \prod_{\alpha \in R_t} (1 - t z^\alpha)} \right)\end{aligned}$$

for some $c_\nu \in \pm q^{\mathbb{Z}} t^{\mathbb{Z}}$.

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- We have a formula for $\nabla \mathcal{G}_\nu$.
- Does there exist formula $\tilde{H}_\mu = \sum_\nu a_{\mu\nu}(q, t) \mathcal{G}_\nu$? Yes!

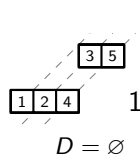
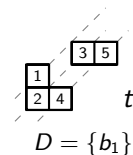
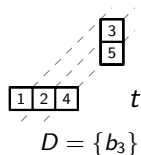
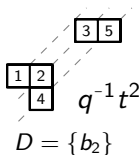
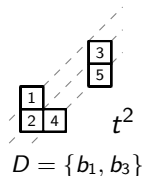
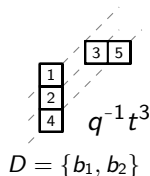
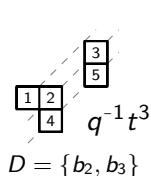
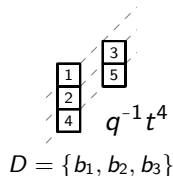
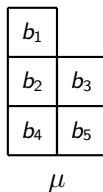
- ① Background on symmetric functions and Macdonald polynomials
- ② Shuffle theorems, combinatorics, and LLT polynomials
- ③ **A new formula for Macdonald polynomials**

Haglund-Haiman-Loehr formula example

$$\tilde{H}_\mu(X; q, t) = \sum_D \left(\prod_{u \in D} q^{-\text{arm}(u)} t^{\text{leg}(u)+1} \right) \mathcal{G}_{\nu(\mu, D)}(X; q)$$

Haglund-Haiman-Loehr formula example

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Putting it all together

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- Take HHL formula $\tilde{H}_\mu = \sum_D a_{\mu,D} \mathcal{G}_{\nu(\mu,D)}$ and apply $\omega \nabla$.
- By construction, all the LLT Catalan animals $H_{\nu(\mu,D)}$ appearing on the RHS will have the same root ideal data (R_q, R_t, R_{qt}) .
- Collect terms to get $\prod_{(b_i, b_j) \in V(\mu)} (1 - q^{\text{arm}(b_i)+1} t^{-\text{leg}(b_i)} z_i/z_j)$ factor for $V(\mu)$ the set of vertical dominoes (b_i, b_j) in μ .

$$\tilde{H}_\mu = \omega \text{pol}_X \sigma \left(z_1 \cdots z_n \frac{\prod_{\substack{\alpha_{ij} \in V(\mu)}} (1 - q^{\text{arm}(b_i)+1} t^{-\text{leg}(b_i)} z_i/z_j) \prod_{\alpha \in \hat{R}_\mu} (1 - qt z^\alpha)}{\prod_{\alpha \in R_+} (1 - qz^\alpha) \prod_{\alpha \in R_\mu} (1 - tz^\alpha)} \right).$$

The root ideal R_μ

b_1		
b_2	b_3	
b_4	b_5	b_6
b_7	b_8	b_9

row reading order

$b_1 \prec b_2 \prec \cdots \prec b_n$

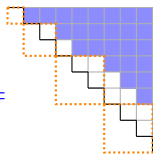
$$R_\mu := \{\alpha_{ij} \in R_+ \mid \text{south}(b_i) \preceq b_j\},$$

$$\hat{R}_\mu := \{\alpha_{ij} \in R_+ \mid \text{south}(b_i) \prec b_j\},$$

$$R_\mu \setminus \hat{R}_\mu \leftrightarrow V(\mu) = \text{vertical dominoes in } \mu$$

Example:

$R_{3321} =$



The root ideal R_μ

b_1		
b_2	b_3	
b_4	b_5	b_6
b_7	b_8	b_9

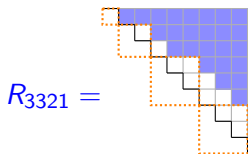
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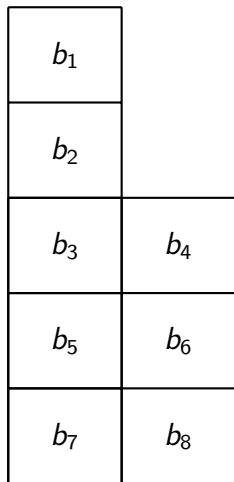
Example:



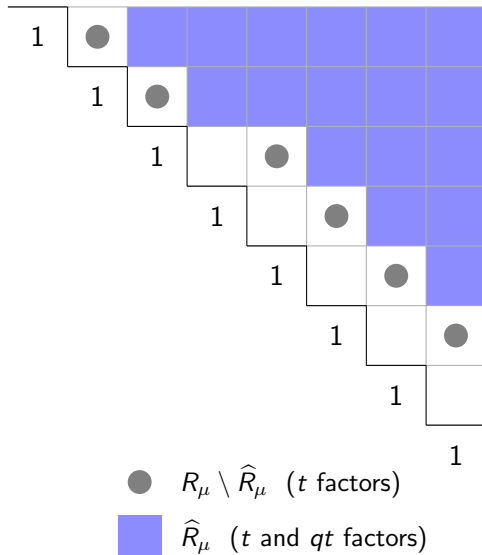
Remark

$$\tilde{H}_\mu(X; 0, t) = \omega \operatorname{pol}_X \sigma \left(\frac{z_1 \cdots z_n}{\prod_{\alpha \in R_\mu} (1 - t z^\alpha)} \right)$$

Example



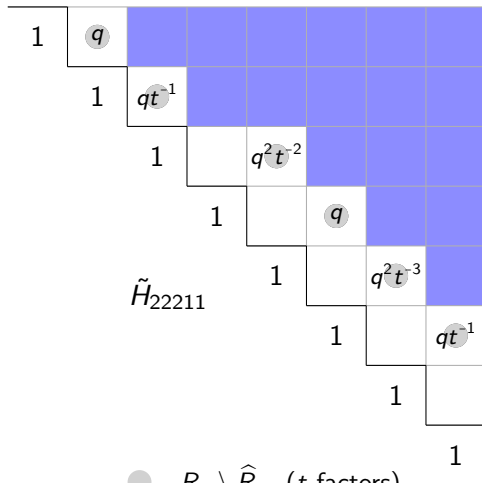
partition $\mu = 22211$



Example

$1 - q^{\frac{z_1}{z_2}}$	
$1 - qt^{-1} \frac{z_2}{z_3}$	
$1 - q^2 t^{-2} \frac{z_3}{z_5}$	$1 - q^{\frac{z_4}{z_6}}$
$1 - q^2 t^{-3} \frac{z_5}{z_7}$	$1 - qt^{-1} \frac{z_6}{z_8}$

numerator factors $1 - q^{\text{arm}+1} t^{-\text{leg}} z_i/z_j$



● $R_\mu \setminus \hat{R}_\mu$ (t factors)

■ $\hat{R}_\mu$ (t and qt factors)

A positivity conjecture

What can this formula tell us that other formulas for Macdonald polynomials do not?

A positivity conjecture

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$$\tilde{H}_\mu^{(s)} = \omega \operatorname{pol}_X \sigma \left((z_1 \cdots z_n)^s \frac{\prod_{\alpha_{ij} \in R_\mu \setminus \hat{R}_\mu} (1 - q^{\operatorname{arm}(b_i)+1} t^{-\operatorname{leg}(b_i)} z_i / z_j) \prod_{\alpha \in \hat{R}_\mu} (1 - q t z^\alpha)}{\prod_{\alpha \in R_+} (1 - q z^\alpha) \prod_{\alpha \in R_\mu} (1 - t z^\alpha)} \right)$$

Conjecture (Blasiak-Haiman-Morse-Pun-S., 2025)

For any partition μ and positive integer s , the symmetric function $\tilde{H}_\mu^{(s)}$ is Schur positive. That is, the coefficients in

$$\tilde{H}_\mu^{(s)} = \sum_{\nu} K_{\nu, \mu}^{(s)}(q, t) s_\nu$$

satisfy $K_{\nu, \mu}^{(s)}(q, t) \in \mathbb{N}[q, t]$.

Symmetric functions, representation theory, and combinatorics

Symmetric function	Representation theory	Combinatorics
s_λ	Irreducible V_λ	SSYT(λ)
$\tilde{H}_\lambda(X; q, t)$	Garsia-Haiman M_λ	HHL
∇e_n	DH_n	Shuffle theorem
$H(R_+, R_+, R_+^0, \mathbf{b})$	??	Generalized shuffle theorem
$\tilde{H}_\lambda^{(s)}(X; q, t)$	??	??

Thank you!

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Schiffmann to Shuffle

- Shuffle algebra S given by the image of Laurent polynomials $\phi \in \mathbb{Q}(q, t)[x_1^{\pm 1}, \dots, x_l^{\pm 1}]$ via map

$$H_{q,t}: \phi \mapsto \sum_{w \in S_l} w \left(\frac{\phi \prod_{i < j} (1 - qtx_i/x_j)}{\prod_{i < j} ((1 - x_j/x_i)(1 - qx_i/x_j)(1 - tx_i/x_j))} \right)$$

- (Schiffmann-Vasserot, 2013) There exists isomorphism $\psi: S \rightarrow \mathcal{E}^+$.
- (Negut, 2014) gives well-defined

$$D_{b_1, \dots, b_l} = \psi \left(H_{q,t} \left(\frac{x_1^{b_1} \cdots x_l^{b_l}}{\prod_{i=1}^{l-1} (1 - qtx_i/x_{i+1})} \right) \right)$$

Key Relationship (BHMPs, 2023)

For $\zeta = \psi(\phi) \in \mathcal{E}^+$,

$$\omega(\zeta \cdot 1) = \omega \operatorname{pol}_X H_{q,t}(\phi).$$

Cauchy Identity

- (Twisted) non-symmetric Hall-Littlewood polynomials $E_{\lambda}^{\sigma}(x; q)$ defined via Demazure-Lusztig operators.

$$T_i = qs_i + (1 - q) \frac{s_i - 1}{1 - x_{i+1}/x_i}$$

- Dual basis F_{λ}^{σ} .

Cauchy identity

$$\frac{\prod_{i < j} (1 - q t x_i y_j)}{\prod_{i \leq j} (1 - t x_i y_j)} = \sum_{\mathbf{a} \geq 0} t^{|\mathbf{a}|} E_{\mathbf{a}}^{\sigma}(x_1, \dots, x_l; q^{-1}) F_{\mathbf{a}}^{\sigma}(y_1, \dots, y_l; q),$$

- $\mathcal{L}_{\beta/\alpha} = H_q(w_0(F_{\beta}^{\sigma^{-1}}(x; q) \overline{E_{\alpha}^{\sigma^{-1}}(x; q)}))$